

LAB#12

BACKUP AND RESTORATION IN MONGODB

OBJECTIVE:

- To learn about backup and restoration
- To implement different methods of backup by using mongodump.
- To implement different methods of restoration by using mongorestore.

THEORY:

Data backup is one of the most highly required processes for any database management system as data can be lost or get corrupted to overcome these drawbacks we need database backup. Database backup is a copy of a database that already exists. In MongoDB, mongodump tool is used to take the data backup. And mongorestore tool is used to restore the backup data.

DATA BACKUP:

In MongoDB, mongodump tool is used to take the data backup. It simply dumps all the data stored into a dump directory of MongoDB. Backed-up data is in BSON format also known as BSON datadumps. By default, the backup is stored in mongodb's bin\dump folder to specify a different output directory we can use the `-out` option. Mongodump is used in two ways with or without argument.

Without Argument:

Without any arguments, mongodump connects with MongoDB instance on the local system on port 27017 and creates a backup of every database and every collection.

- 1) Create a folder with name mongodb on desktop.
- 2) Now open command prompt in this folder by writing `cmd` in the path
- 3) Write `mongodump` in the command prompt. Here, you will see that it is not recognized.
- 4) Download and install 'MongoDb command line tools' on the following link and select the 'msi' package
<https://www.mongodb.com/try/download/database-tools>

MongoDB Command Line Database Tools Download

The MongoDB Database Tools are a collection of command-line utilities for working with a MongoDB deployment. These tools release independently from the MongoDB Server schedule enabling you to receive more frequent updates and leverage new features as soon as they are available. See the [MongoDB Database Tools](#) documentation for more information.

Version	100.9.4	▼
Platform	Windows x86_64	▼
Package	msi	▼

- 5) Set the environment variables as we did in the installation of MongoShell.
- 6) Now write Mongodump in the command line of mongodb folder. It will output like following

Mongodump

```
C:\Users\Furqan\Desktop\mongodb>mongodump
2024-01-14T23:14:53.345+0500   writing admin.system.version to dump\admin\system.version.bson
2024-01-14T23:14:53.347+0500   done dumping admin.system.version (1 document)
2024-01-14T23:14:53.347+0500   writing Books.collection to dump\Books\collection.bson
2024-01-14T23:14:53.348+0500   writing FirstDB.NewColl to dump\FirstDB\NewColl.bson
2024-01-14T23:14:53.349+0500   writing FirstDB.products to dump\FirstDB\products.bson
2024-01-14T23:14:53.349+0500   writing Books.authors to dump\Books\authors.bson
2024-01-14T23:14:53.381+0500   done dumping FirstDB.NewColl (10 documents)
2024-01-14T23:14:53.382+0500   done dumping Books.collection (1 document)
2024-01-14T23:14:53.400+0500   done dumping FirstDB.products (12 documents)
2024-01-14T23:14:53.400+0500   done dumping Books.authors (2 documents)
2024-01-14T23:14:53.400+0500   writing Books.Authors to dump\Books\Authors.bson
2024-01-14T23:14:53.400+0500   writing FirstDB.ProductsQty to dump\FirstDB\ProductsQty.bson
2024-01-14T23:14:53.402+0500   done dumping Books.Authors (0 documents)
2024-01-14T23:14:53.435+0500   done dumping FirstDB.ProductsQty (0 documents)
```

Here, you can see that all the databases has been copied in the mongodb's dump folder

This PC > Desktop > mongodb > dump			
	Name	Date modified	Type
s	admin	14/01/2024 11:14 PM	File folder
s	Books	14/01/2024 11:14 PM	File folder
s	FirstDB	14/01/2024 11:14 PM	File folder

With Argument:

By specifying the database in the argument by which we can limit the amount of data stored in the database dump

mongodump --db databaseName --collection collectionName

To specify a different output directory we can use the `--out` option:

Example#1:

Here we have created a backup for databases 'FirstDB' by using `-d` in the specified folder name 'FirstDBbackup' by using `--out`.

```
C:\Users\Furqan\Desktop\mongodb>mongodump -d FirstDB --out FirstDBbackup
2024-01-14T23:26:56.888+0500   writing FirstDB.ProductsQty to FirstDBbackup\FirstDB\ProductsQty.bson
2024-01-14T23:26:56.889+0500   writing FirstDB.NewColl to FirstDBbackup\FirstDB\NewColl.bson
2024-01-14T23:26:56.910+0500   writing FirstDB.products to FirstDBbackup\FirstDB\products.bson
2024-01-14T23:26:56.910+0500   done dumping FirstDB.ProductsQty (0 documents)
2024-01-14T23:26:56.931+0500   done dumping FirstDB.NewColl (10 documents)
2024-01-14T23:26:56.932+0500   done dumping FirstDB.products (12 documents)
```

Example#2:

Here, we have specified backup for single collection with the name 'products' by using `-c` with the collection's name.

```
C:\Users\Furqan\Desktop\mongodb>mongodump -d FirstDB -c products --out productsCollection
2024-01-14T23:33:44.418+0500   writing FirstDB.products to productsCollection\FirstDB\products.bson
2024-01-14T23:33:44.420+0500   done dumping FirstDB.products (12 documents)
```

Now, let's drop the database and try to restore it.

```
> use FirstDB
< switched to db FirstDB
> db.dropDatabase()
< { ok: 1, dropped: 'FirstDB' }
```

DATA RESTORATION

In MongoDB, `mongorestore` utility is used to restore the backup data. It restores the binary backup created by `mongodump` utility (i.e., BSON data dumps). It can restore either an entire database backup or a subset of the backup. It also restores the indexes which are created for any collection inside that database. By default, `mongorestore` looks for a database backup in `mongodb's bin\dump` folder which is also the default folder for `mongodump` command for dumping the backup.

```
C:\Users\Furqan\Desktop\mongodb>mongorestore
2024-01-14T23:47:14.294+0500 using default 'dump' directory
2024-01-14T23:47:14.295+0500 preparing collections to restore from
2024-01-14T23:47:14.298+0500 reading metadata for Books.authors from dump\Books\authors.metadata.json
2024-01-14T23:47:14.299+0500 reading metadata for Books.collection from dump\Books\collection.metadata.json
2024-01-14T23:47:14.299+0500 reading metadata for FirstDB.NewColl from dump\FirstDB\NewColl.metadata.json
2024-01-14T23:47:14.300+0500 reading metadata for FirstDB.ProductsQty from dump\FirstDB\ProductsQty.metadata.json
2024-01-14T23:47:14.301+0500 reading metadata for FirstDB.products from dump\FirstDB\products.metadata.json
2024-01-14T23:47:14.458+0500 restoring to existing collection Books.collection without dropping
2024-01-14T23:47:14.458+0500 restoring Books.collection from dump\Books\collection.bson
2024-01-14T23:47:14.460+0500 restoring to existing collection Books.authors without dropping
```

```
>_MONGOSH
> show dbs
< Books    120.00 KiB
  admin    40.00 KiB
  config   108.00 KiB
  local    80.00 KiB
> show dbs
< Books    136.00 KiB
  FirstDB   148.00 KiB
```

Here, you can see the before and after of database restoration display.

Example#2

Restoring single database backup folder

```
C:\Users\Furqan\Desktop\mongodb>mongorestore FirstDBbackup
2024-01-15T00:00:10.178+0500 preparing collections to restore from
2024-01-15T00:00:10.179+0500 reading metadata for FirstDB.NewColl from FirstDBbackup\FirstDB\NewColl.metadata.json
2024-01-15T00:00:10.180+0500 reading metadata for FirstDB.ProductsQty from FirstDBbackup\FirstDB\ProductsQty.metadata.json
```

Example#3

Restoring specified collection from database.

```
C:\Users\Furqan\Desktop\mongodb>mongorestore --nsInclude FirstDB.products productsCollection
2024-01-15T00:10:05.755+0500 preparing collections to restore from
2024-01-15T00:10:05.757+0500 reading metadata for FirstDB.products from productsCollection\FirstDB\products.metadata.json
2024-01-15T00:10:05.758+0500 restoring to existing collection FirstDB.products without dropping
2024-01-15T00:10:05.758+0500 restoring FirstDB.products from productsCollection\FirstDB\products.bson
```

LAB TASKS:

- 1) Create a backup for all the databases.
- 2) Create a backup for the specified database
- 3) Delete the specified database and restore it.
- 4) Explore the queries and process for backup and restoration in SQL.

